PennState Extension

Pepper Production

Peppers lend themselves well to small-scale and part-time farming operations.

Updated: October 27, 2020



Peppers

Peppers lend themselves well to small-scale and part-time farming operations. Various markets exist for growers with small-acreage farms (fewer than 5 acres), and the various mature fruit colors (green, red, yellow, orange, purple, and brown), shapes, and heat levels (from sweet to extra hot) make it easier for growers to find niche markets. Many field operations, such as land preparation, planting, and harvesting, can be custom hired, and any equipment owned by the grower can be used for most other vegetable crops.

Peppers (*Capsicum annuum* and *Capsicum frutescens*), both sweet and hot, originated in southern Mexico and Central and South America. Evidence of their cultivation can be traced back thousands of years. Christopher Columbus found peppers growing in the West Indies, but they were not introduced in Europe until the sixteenth century. South America, Spain, England, and the Caribbean were involved in their introduction to North America. According to U.S. Department of Agriculture records, commercial bell peppers were first produced in the southern United States in 1925. Today, seed companies distribute several hundred varieties of both sweet and hot peppers.

Most of the peppers harvested in the United States are sold as fresh produce. The United States produces over 70,000 acres of peppers, over one-third of which are grown for the fresh market. In 2017, the Northeast produced over 7,300 acres of all types of peppers on 6,500 farms (2017 Census of Agriculture).

Marketing

Fresh-market peppers are produced in Pennsylvania from early June to the end of October. Pepper cultivars recommended for Pennsylvania are listed in Table 1. Fresh-market peppers are usually sold loose in bulk containers. Several basic marketing alternatives are available to the pepper grower: wholesale markets, cooperatives, local retailers (grocery stores), roadside stands, farmers markets, community-supported agriculture (CSA), pick-your-own operations, and processing firms. When planning production, first consider your ability and access to market. You should conduct some market research—growers often overestimate their ability to sell in a given market. Production of less than one acre of many vegetable crops is typical for most growers.

In wholesale marketing, producers often contract with shippers to market and ship the peppers for a predetermined price. If you do not use a contractor and ship your peppers to a wholesale market yourself, your product will be subject to the greatest price fluctuations. Produce auctions are an outlet that operate weekly; however, you must deliver the peppers to the auction. Marketing cooperatives generally use a daily pooled cost and price, which spreads price fluctuations over all participating producers. Local retailers are another possible market, but you must take the time to contact produce managers and provide good-quality peppers when stores require them. Depending on your location, processors may or may not be a marketing option. Processors are less likely to contract with small-acreage growers (those with fewer than 5 acres). For more information on marketing, consult "Agricultural Alternatives: Fruit and Vegetable Marketing for Small-Scale and Part-Time Growers."

Table 1. Recommended pepper cultivars for Pennsylvania.

Name	Color
Bell	
Aristotle (BLSR 1-3, PHY, TM)	Green to red
Declaration (BLSR 1-3, 5, PHY, TSWV)	Green to red
Delerio (TMV, TSWV)	Green to orange
Early Sunsation (BLSR 1-3)	Green to yellow
Intruder (BLSR 1-3, PHY, TEV, TMV)	Green to red
Karisma (BLSR 1-3, CMV, PVY, TMV)	Green to red
Mecate (BLSR 1-3, TMV)	Green to yellow
Mercer (BLSR 0-3, 7, 8, PHY, TMV)	Green to red
Paladin (PHY, TM)	Green to red
Playmaker (BLSR 0-10, PHY, TM)	Green to red
Red Knight (BLSR 1-3, PVY)	Green to red
Revolution (BLSR 1-3, 5, CMV, PHY)	Green to red
Turnpike (BLSR 0-5, 7-9, PHY)	Green to red

Name	Color
1819 (BLSR 1-5, PHY)	Green to red
Cherry	
Fireball	Green to red
Grandi	Green to red
Supper Sweet Cherry	Green to red
Sweet Frying	
Aruba (PHY)	Light green
Biscayne	Light yellow
Carmen	Green to red
Cheyenne (Cayenne)	Green to red
Key West (BLSR 1-3)	Light green to red
Red Crest	Green to red
Yellow Crest	Green to yellow
Hot	
Campeon (Jalapeño; BLSR 0-3, 7, 8, PVY)	Green to red
Compadre (Jalapeño)	Green to red
El Jefe (Jalapeño; BLSR 0-3, 7, 8, PVY, TEV)	Green to red
Grande (Jalapeño; PVY, TEV)	Green to red
Mesilla (Cayenne; PVY, TEV, TMV)	Green to red
Nainari (Cayenne)	Green to red
New Park (Jalapeño; BLSR 1-3)	Green to red
Numex Joe. E. Parker (Anaheim)	Green to red
Rayo (Jalapeño; BLSR 1-3)	Green to red
Banana and Hungarian	
Boris (TSWV)	Yellow to red

Name	Color
Bounty	Yellow to red
Budapest (hot)	Yellow to red
Doblon (TMV, TSWV)	Yellow to red
Ethem	Yellow to red
Goldrush (BLSR 2)	Yellow to red
Inferno (hot)	Yellow to red
Pageant (BLSR 1-3)	Yellow to red
Sopron (BLSR 1-3)	Yellow to red
Sweet Savannah	Yellow to red
Sweet Sunset (BLSR 1-3)	Yellow to red

Cultivars are hybrids and listed in alphabetical order within type.
BLSR = resistant to bacterial leaf spot
CMV = tolerant to cucumber mosaic virus
PVY = resistant to potato virus Y
TEV = resistant or tolerant to tobacco etch virus
TM = resistant to tobacco mosaic virus
TSWV = resistant or tolerant to tomato spotted wilt virus
PHY = resistant or tolerant to Phytophthora

Retail marketing options include roadside stands (either your own or another grower's), farmers markets, CSA, and pick-your-own operations. These options can provide opportunities to receive higher-than-wholesale prices for your peppers, but you may have some additional expenses for advertising, building and maintaining a facility, and providing service to your customers. With pick-your-own operations, you save on harvest costs, but you must be willing to accept some waste. Farmers markets are another retail option, but you should contact the markets well in advance of the marketing season to be sure space is available and to find out what requirements you must follow. For more information about roadside markets, see "Agricultural Alternatives: Developing a Roadside Farm Market (https://extension.psu.edu/developing-a-roadside-farm-market)."

Production Considerations

Peppers grow best on well-drained soils that have good waterholding characteristics and a pH of 5.8 to 6.6. Peppers are started as transplants in the greenhouse six to eight weeks prior to planting in the field. Because peppers are a warmseason crop, they should not be transplanted until the soil temperature 3 inches beneath the soil surface reaches 60°F. Peppers grow well on raised beds covered with black or silver plastic mulch. Providing the plants with drip irrigation ensures optimum plant growth and yields and provides the option to apply injection-based fertilizer during the growing season. For more information on drip irrigation, consult "Agricultural Alternatives: Drip Irrigation for Vegetable Production(https://extension.psu.edu/drip-irrigation-for-vegetable-production)."

Growers generally plant approximately 10,000 to 14,000 plants per acre in double rows spaced 14 to 18 inches apart on plastic mulched beds with 16 to 24 inches between plants in the row and with the beds spaced 5 to 6.5 feet apart from their centers. A single row of peppers can also be planted on each bed (5,000 to 6,500 plants per acre). Staking may be needed, depending on cultivar and cultivation methods.

Phosphorus, potassium, and lime rates should be based on annual soil test results. Applying 120 pounds of nitrogen per acre is recommended. For high-clay soils high in Pennsylvania, a general recommendation is to apply a portion of the nitrogen (40 to 60 percent) prior to planting and the remainder throughout the growing season via the drip irrigation system.

Pest Management

Weed control can be achieved with herbicides, plastic mulch, and a good crop-rotation system. Several preplant and postemergence herbicides are available for peppers, depending on the specific weed problem and pepper growth stage. If infestation levels are mild, early cultivation can minimize weed problems.

Insects are a major problem in pepper production. Aphids, flea beetles, pepper maggots, thrips, and European corn borers can all cause crop losses. Monitoring insect populations with traps and scouting will help you determine when you should use pesticides and how often you should spray.

Several pepper diseases can cause crop losses, including bacterial leaf spot, phytophthora blight, anthracnose fruit rot, and bacterial soft rot, and viruses such as potato virus Y and tobacco mosaic virus. These diseases can be managed by using disease-resistant and disease-tolerant cultivars, having a good crop rotation plan, growing in locations with good air movement, and planting in soils with good water infiltration.

Many of the pesticides necessary for pepper production are restricted-use pesticides and require a pesticide license to purchase. Pesticide applicators tests are usually administered at county extension offices; contact your local office for dates and times. When using any pesticides in your enterprise, you must follow all label recommendations regarding application rates and personal protection equipment (PPE) requirements. Also remember that worker protection standards (WPS) apply to both owner and employees.

Harvest and Storage

Most peppers are harvested by hand two to four times during the growing season. Mechanical harvesters are available for harvesting hot peppers (jalapeño, chiles, and hot cherries) with a once-over harvest. You will need to grade peppers for size and color and check for worms and insect damage to ensure you are marketing a high-quality product.

Cooling the peppers after harvest will remove field heat, which improves shelf life. You should refrigerate the peppers immediately after harvest to maintain quality. Peppers will retain good quality for approximately 14 to 21 days if stored at 90 to 95 percent humidity and 47 to 55°F.

Environmental Regulations

All agricultural operations in Pennsylvania, including small-scale and part-time farming enterprises, operate under the Pennsylvania Clean Streams Law. A specific part of this law is the Nutrient Management Act. Portions of the act may or may not pertain to your operation, depending on whether you have livestock on your farm. However, all operations may be a source of surface water or groundwater pollution. Because of this possibility, you should contact your local Soil and Water Conservation District to determine what regulations may pertain to your operation.

Good Agricultural Practices and Good Handling Practices

Good Agricultural Practices (GAP) and Good Handling Practices (GHP) are voluntary programs that you may wish to consider for your operation. The idea behind these programs is to ensure a safer food system by reducing the chances for foodborne illnesses resulting from contaminated products reaching consumers. Several major food distribution chains are beginning to require GAP- and GHP-certified products from their producers. These programs set standards for worker hygiene, use of manure, and water supply quality.

These practices require an inspection from a designated third party, and there are fees associated with the inspection. Prior to an inspection, you will need to develop and implement a food safety plan and designate someone in your operation to oversee this plan. You will need to have any water supply used by your workers or for crop irrigation and pesticide application checked at least twice each year. A checklist of the questions to be asked during the inspection(https://www.ams.usda.gov/services/auditing/gap-ghp/audit) can be found online. For more information about GAP and GHP, contact your local extension office or state department of agriculture.

You may also be impacted by the Food Safety and Modernization Act (FSMA), administered by the Food and Drug Administration (FDA). FSMA governs record keeping, health and hygiene, water quality, and animal waste used as fertilizer, among additional requirements. For more information about FSMA, visit Penn State Extension's FSMA article (https://extension.psu.edu/food-safety-modernization-act-produce-safety-rule).

Risk Management

You should carefully consider how to manage risk on your farm. First, you should insure your facilities and equipment. This may be accomplished by consulting your insurance agent or broker. It is especially important to have adequate levels of property, vehicle, and liability insurance. You will also need workers compensation insurance if you have any employees. You may also want to consider your needs for life and health insurance and if you need coverage for business interruption or employee dishonesty. For more on agricultural business insurance, see "Agricultural Alternatives: Agricultural Business Insurance (https://extension.psu.edu/agricultural-business-insurance)."

Second, check to see if there are multi-peril crop insurance programs available for your crop or livestock enterprises. There are crop insurance programs designed to help farmers manage both yield risk and revenue shortfalls. However, individual crop insurance coverage is not available for all crops. If individual coverage is not available for what you grow, you may be able to use the Whole Farm Revenue Protection (WFRP) program to insure the revenue of your entire farm operation. Information from your Schedule F tax records (or a "Substitute Schedule F for WFRP Purposes" if you do not file a Schedule F) from the past five consecutive years is used to calculate the WFRP policy's approved revenue guarantee. Operations that have expanded over time may be allowed to increase the approved revenue amount based on an indexing procedure. Depending on the number of commodities grown, you have the choice of coverage of 50 to 85 percent of your approved revenue. Coverage and premium costs depend on the level of diversification in your operation; the maximum level of insured revenue is $8.5 million (based on maximum adjusted gross revenues of $17 million and the 50 percent coverage level). WFRP also provides replant coverage if it not already covered under an underlying individual crop policy. More information on WFRP (https://www.rma.usda.gov/en/Policy-and-Procedure/Insurance-Plans/Whole-Farm-Revenue-Protection) can be found online.

Finally, the USDA Farm Service Agency has a program called the Noninsured Assistance Program (NAP) that is designed to provide a minimal level of yield risk protection for producers of commercial agricultural products that don't have multi-peril crop insurance coverage. NAP is designed to reduce financial losses when natural disasters cause catastrophic reduction in production. A basic level of coverage (50 percent of expected production at 55 percent of the average market price) is available for a fee of $325 per crop per county (fees are capped at $825 per producer per county, but not to exceed a total of $1,950 for producers growing crops in multiple counties). Higher levels of protection at the 50, 55, 60, and 65 percent levels at 100 percent of the average market price are available for additional premium. NAP coverage is available through your local USDA Farm Service Agency office. The application fee for this program may be waived for eligible limited-resource farmers.

Sample Budget

Included in this publication is a sample fresh-market pepper production budget (on page 6). This budget utilizes custom hire for most of the field work, which could be more economical for smaller acreages. If you have your own equipment, substitute your costs for the custom-hire costs. The budget summarizes the receipts, costs, and net returns of a pepper enterprise. This sample budget should help ensure that all costs and receipts are included in your calculations. Costs and returns are often difficult to estimate in budget preparation because they are numerous and variable. Therefore, you should think of this budget as an approximation and make appropriate adjustments in the "Your Estimate" column to reflect your specific production and resource situation. These budgets are developed for one acre; however, your scale of production should be based on your market considerations. More information on the use of crop budgets can be found in "Agricultural Alternatives: Budgeting for Agricultural Decision Making(https://extension.psu.edu/budgeting-for-agricultural-decision-making)."

You can make changes to the interactive PDF budget files for this publication by inputting your own prices and quantities in the green outlined cells for any item. The cells outlined in red automatically calculate your revised totals based on the changes you made to the cells outlined in green. You will need to click on and add your own estimated price and quantity information to all of the green outlined cells to complete your customized budget. When you are done, you can print the budget using the green Print Form button at the bottom of the form. You can use the red Clear Form button to clear all the information from your budget when you are finished.

Sample Budget Worksheet

- Sample Pepper Budget-Fresh Market-Bell Type

Initial resource requirements for fresh market bell peppers

Land: 1 acre

Labor:

- Production labor: 12-20 hours
- Harvest labor: 125-175 hours
- Grading and packing labor: 30-35 hours

Capital: $10,500-$12,500

Equipment:

- Tractor 40-75 hp
- Vegetable transplanter
- Plastic mulch layer
- Boom sprayer

For More Information

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This publication was developed by the Small-scale and Part-time Farming Project at Penn State with support from the U.S. Department of Agriculture-Extension Service.

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